



12. India's target under Paris Agreement is to reduce emission intensity by:
 A) 10–15% by 2020 B) 20–25% by 2020
 C) 33–35% by 2030 D) 50% by 2030
13. Which risk management tool is preventive?
 A) Emergency evacuation plan B) Environmental regulations
 C) Insurance claims D) Disaster compensation
14. Risk perception refers to:
 A) Public understanding of risks B) Government policy only
 C) Laboratory experiments D) Legal enforcement only
15. Which organization develops global guidelines on chemical risk assessment?
 A) WHO B) UNESCO
 C) WTO D) IMF
16. Environmental risk management aims at:
 A) Profit maximization B) Reducing adverse impacts on health and ecosystems
 C) Improving trade only D) Weakening regulations
17. Quantitative exposure models are used to calculate:
 A) Market risk B) Hazard probability
 C) Political influence D) Dose received by population
18. Which parameter is essential in calculating flood risk?
 A) Rainfall intensity B) River discharge
 C) Vulnerability of settlement D) All of the above
19. Which measure indicates annual probability of hazard occurrence?
 A) Annual Exceedance Probability (AEP) B) Return Period
 C) Cumulative Frequency D) Risk Index
20. In probabilistic risk assessment, uncertainty is handled by:
 A) Sensitivity analysis B) Mathematical modeling
 C) Probability distributions D) All of the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the need and importance of environmental legislation in India	08	L*2	1	2,7
	b) A factory is discharging effluents into a river. Explain the applicable laws and the jurisdiction of enforcement agencies.	08	L2	2	2
2.	a) Explain the powers and functions of the Central Pollution Control Board (CPCB).	08	L2	1	1,7
	b) A hospital fails to segregate biomedical waste. Discuss the case in the light of Biomedical Waste Management Rules, 2016.	08	L2	2	1,2
3.	a) Describe the scope and significance of the Water (Prevention and Control of Pollution) Act, 1974.	08	L2	1	2,7
	b) A pesticide manufacturer imports banned chemicals. Which law is violated? Justify your answer.	08	L2	2	1,2
Unit – II					
4.	a) Define sustainable development and explain its importance in environmental policy.	08	L2	3	2.4
	b) Define risk assessment and explain its importance in environmental management.	08	L2	4	1,7
5.	a) Assess the effectiveness of the Wildlife Protection Act, 1972 in conserving biodiversity.	08	L2	3	1,2

b)	Discuss the role of hazard identification in risk assessment.	08	L2	4	1,2
6. a)	Examine the adequacy of the Air Act, 1981 in addressing current air pollution challenges.	08	L2	3	1,2
b)	Discuss the importance of dose-response assessment in chemical risk evaluation.	08	L2	4	2

Unit – III

7. a)	Define Quantitative Risk Assessment (QRA) and explain its importance in environmental management.	08	L2	5	1,7
b)	Explain the mathematical formulation of risk as the product of Probability and Consequence; support your answer with suitable examples.	08	L2	5	2
8. a)	Explain the steps involved in developing a QRA framework.	08	L2	5	2,7
b)	Explain the role of hazard identification in QRA.	08	L2	5	2,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
